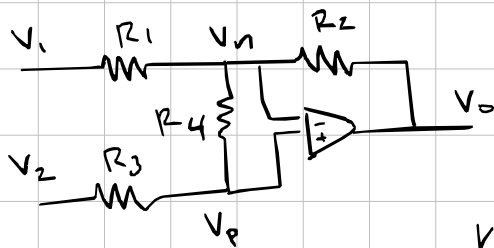


Q1 i) Ideal V_o wrt. V_1, V_2 . $r_d = \infty$ $r_o = 0$

↳ use virtual short

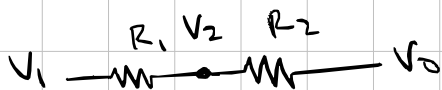


V_n must be = to V_p

therefore: $i_4 = 0$

KCL @ V_p : $i_3 = i_4 \rightarrow i_3 = 0$

therefore $V_n = V_p = V_2$

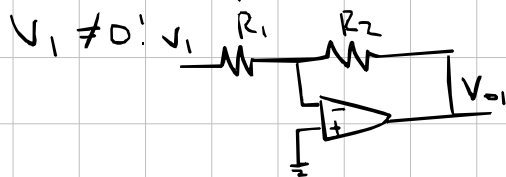


$$V_2 = (V_o - V_1) \frac{R_1}{R_1 + R_2} + V_1$$

$$\begin{aligned} \frac{R_1}{R_1 + R_2} V_o &= V_2 + \left(\frac{R_1}{R_1 + R_2} - 1 \right) V_1 \\ &= V_2 - \frac{R_2}{R_1 + R_2} V_1 \end{aligned}$$

$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_2 - \frac{R_2}{R_1} V_1$$

Via Superposition knowing $V_p = V_2$ & $i_4 = 0$



$$V_{o1} = -\frac{R_2}{R_1} V_1$$

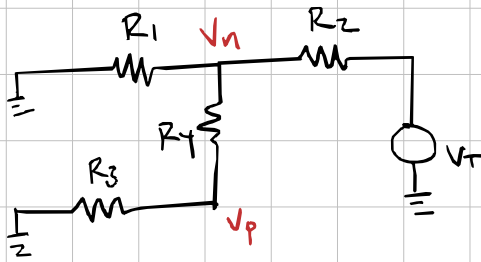
$V_2 \neq 0$:



$$V_{o2} = \left(1 + \frac{R_2}{R_1} \right) V_2$$

$$V_o = -\frac{R_2}{R_1} V_1 + \left(1 + \frac{R_2}{R_1} \right) V_2$$

ii)



$$V_p = \frac{R_3}{R_3 + R_4} V_n$$

$$V_n = \frac{R_1 \parallel (R_3 + R_4)}{R_1 \parallel (R_3 + R_4) + R_2} V_T$$

$$V_D = V_p - V_n$$

$$V_D = \left(\frac{R_3}{R_3 + R_4} - 1 \right) V_n$$

$$= \frac{R_1(R_3 + R_4) / (R_1 + R_3 + R_4)}{\left(R_1(R_3 + R_4) / (R_1 + R_3 + R_4) \right) + R_2} V_T$$

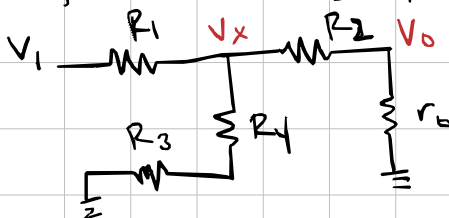
$$V_D = -\frac{R_4}{R_3 + R_4} V_n$$

$$= \frac{R_1(R_3 + R_4)}{R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4)} V_T$$

$$V_D = -\frac{R_1 R_4}{R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4)} V_T$$

$$T = a R_1 R_4 / R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4)$$

iii) aft: with $V_2 = 0$, $r_o \neq 0$



$$V_o = \frac{r_o}{R_2 + r_o} V_x$$

$$V_x = \frac{(R_3 + R_4) \parallel (R_2 + r_o)}{(R_3 + R_4) \parallel (R_2 + r_o) + R_1} V_1$$

$$\frac{V_o}{V_1} = a_{ft} = \left[\frac{r_o}{R_2 + r_o} \right] \left[\frac{(R_3 + R_4)(R_2 + r_o) / (R_2 + R_3 + R_4 + r_o)}{\left((R_3 + R_4)(R_2 + r_o) / (R_2 + R_3 + R_4 + r_o) \right) + R_1} \right]$$

$$a_{ft} = \frac{r_o(R_3 + R_4)}{(R_3 + R_4)(R_2 + r_o) + R_1(R_2 + R_3 + R_4 + r_o)}$$

2) 4050

$$V_2 = 0, r_0 \neq 0 \rightarrow A = \frac{V_0}{V_1}$$

$$A = A_{ideal} / (1 + 1/T) + a_{gt} / (1 + T)$$

$$V_0 = \left(1 + \frac{R_2}{R_1}\right) V_2 - \frac{R_2}{R_1} V_1$$

0

$$A_{ideal} = -\frac{R_2}{R_1}$$

$$A = -\frac{R_2}{R_1} \left/ \left[1 + \frac{(R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4))}{a R_1 R_4} \right] \right.$$

$$\left. + \frac{r_0(R_3 + R_4) / ((R_3 + R_4)(R_2 + r_0) + R_1(R_2 + R_3 + R_4 + r_0))}{1 + a R_1 R_4 / (R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4))} \right]$$

Note! there should be a $r_0 \neq 0$ in the T term, but omitted when calculated in ii). Since $r_0 \ll R_2$ (probably) then OK.

6050

with $r_0 \rightarrow 0$, then $a_{gt} = 0$

if $V_1 \neq 0, V_2 = 0$: $A_1 = \frac{V_{01}}{V_1} = A_{ideal1} / (1 + 1/T) \rightarrow V_{01} = A_{ideal1} / (1 + 1/T) V_1$

if $V_2 \neq 0, V_1 = 0$: $A_2 = \frac{V_{02}}{V_2} \Rightarrow V_{02} = A_{ideal2} / (1 + 1/T) V_2$

$$V_0 = A_{ideal1} / (1 + 1/T) V_1 + A_{ideal2} / (1 + 1/T) V_2 = \frac{1}{1 + 1/T} \underbrace{[A_{ideal1} V_1 + A_{ideal2} V_2]}_{\text{ideal } V_0}$$

$$V_0 = \left[\left(1 + \frac{R_2}{R_1}\right) V_2 - \frac{R_2}{R_1} V_1 \right] \left/ \left[1 + \frac{(R_1(R_3 + R_4) + R_2(R_1 + R_3 + R_4))}{a R_1 R_4} \right] \right.$$

Note! T doesn't depend on V_1 or V_2 !

$$v) \quad V_o = \left(1 + \frac{R_2}{R_1}\right) V_2 - \frac{R_2}{R_1} V_1$$

$$\text{let } V_2 = V_{cm} + V_{dm}/2$$

$$V_1 = V_{cm} - V_{dm}/2$$

$$\begin{aligned} V_o &= \left(1 + \frac{R_2}{R_1}\right) \left(V_{cm} + \frac{V_{dm}}{2}\right) - \frac{R_2}{R_1} \left(V_{cm} - \frac{V_{dm}}{2}\right) \\ &= V_{cm} + \frac{V_{dm}}{2} + \cancel{\frac{R_2}{R_1} V_{cm}} + \frac{R_2}{R_1} \frac{V_{dm}}{2} - \cancel{\frac{R_2}{R_1} V_{cm}} + \frac{R_2}{R_1} \frac{V_{dm}}{2} \end{aligned}$$

$$V_o = \underbrace{\left(\frac{1}{2} + \frac{R_2}{R_1}\right)}_{A_{dm}} V_{dm} + V_{cm} \quad \hookrightarrow A_{cm} = 1$$

$$A_{dm} = \frac{1}{2} + \frac{R_2}{R_1} \quad A_{cm} = 1$$

$$vi) \quad CMRR_{dB} = 20 \log_{10} \left(\frac{A_{dm}}{A_{cm}} \right) = 20 \log_{10} \left(\frac{1}{2} + \frac{R_2}{R_1} \right)$$

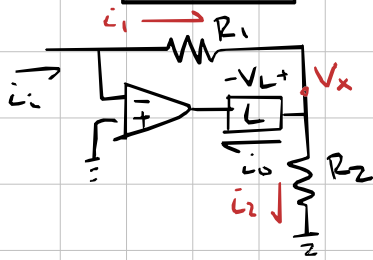
$$40 \text{ dB} = 20 \log_{10} \left(\frac{1}{2} + \frac{R_2}{R_1} \right)$$

$$100 = \frac{1}{2} + \frac{R_2}{R_1} \rightarrow \frac{R_2}{R_1} = 99.5$$

$$\text{let } R_1 = 1 \text{ k}\Omega, \text{ then } R_2 = 99.5 \text{ k}\Omega$$

Q2 i)

4050

 $r_d = \infty, r_o = 0, a = \infty \rightarrow$ virtual short.

$$V_n = V_p = 0 \quad \bar{i}_i = \bar{i}_1$$

$$V_n - V_x = R_1 \bar{i}_1 = R_1 \bar{i}_i = -V_x$$

$$V_x - 0 = R_2 \bar{i}_2 \rightarrow \bar{i}_2 = V_x / R_2$$

$$\bar{i}_1 = \bar{i}_o + \bar{i}_2$$

$$\bar{i}_i = \bar{i}_o + V_x / R_2 = \bar{i}_o - \frac{R_1}{R_2} \bar{i}_i$$

$$\bar{i}_o = \left(1 + \frac{R_1}{R_2}\right) \bar{i}_i \quad \text{where } A = 1 + \frac{R_1}{R_2}$$

$$R_o = \infty$$

6050

$$V_n - V_x = R_1 \bar{i}_1 = R_1 \bar{i}_i \quad *$$

$$V_x - 0 = R_2 \bar{i}_2 \rightarrow \bar{i}_2 = V_x / R_2$$

$$V_o = a(V_p - V_n) = -aV_n$$

$$V_x = V_o + V_L = -aV_n + V_L \rightarrow \bar{i}_2 = \frac{-aV_n + V_L}{R_2}$$

$$\bar{i}_1 = \bar{i}_o + \bar{i}_2 \rightarrow \bar{i}_o = \bar{i}_i - \frac{-aV_n + V_L}{R_2}$$

$$*: V_n + aV_n - V_L = R_1 \bar{i}_i$$

$$V_n(1+a) = \bar{i}_i R_1 + V_L \rightarrow V_n = \frac{\bar{i}_i R_1 + V_L}{1+a}$$

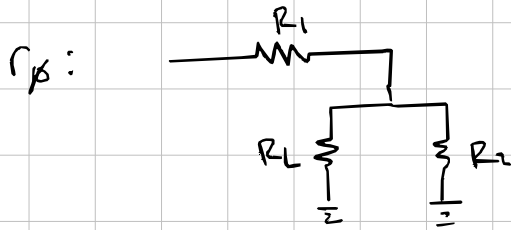
$$\bar{i}_o = \bar{i}_i + \frac{a \bar{i}_i R_1}{R_2(1+a)} + \frac{a V_L}{R_2(1+a)} - \frac{V_L}{R_2}$$

$$\bar{i}_o = \left(1 + \frac{R_1}{R_2} \frac{a}{1+a}\right) \bar{i}_i + \frac{1}{R_2} \left(\frac{a}{1+a} - 1\right) V_L$$

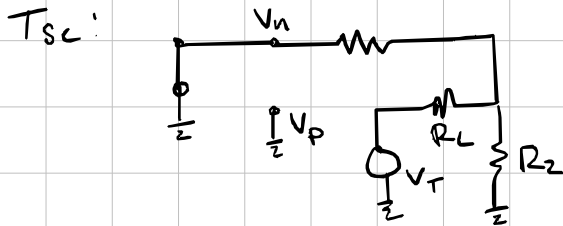
$$\bar{i}_o = \left(1 + \frac{R_1}{R_2} \frac{a}{1+a}\right) \bar{i}_i + \frac{1}{R_2} \frac{1}{1+a} V_L$$

$$A = \left(1 + \frac{R_1}{R_2} \frac{a}{1+a}\right) \quad R_o = R_2(1+a)$$

ii) 4050 calculate R_i with $a \neq \infty$

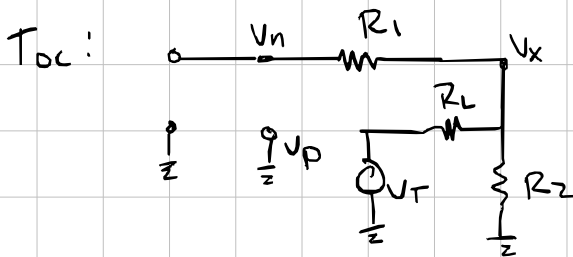


$$r_p = R_1 + R_2 \parallel R_L = R_1 + \frac{R_2 R_L}{R_2 + R_L}$$



$$V_n = V_p = 0$$

$$T_{sc} = 0$$



No current through R_1

$$\therefore V_n = V_x$$

$$V_n = V_T \frac{R_2}{R_2 + R_L}$$

$$T_{oc} = -a \frac{(0 - V_T \frac{R_2}{R_2 + R_L})}{V_T}$$

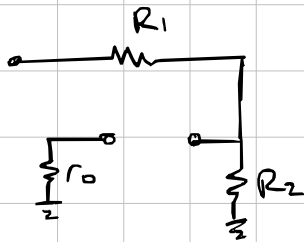
$$T_{oc} = a \frac{R_2}{R_2 + R_L}$$

$$R_i = \left[R_1 + \frac{R_2 R_L}{R_2 + R_L} \right] \left[\frac{1}{1 + a \frac{R_2}{R_2 + R_L}} \right]$$

$$R_i = \frac{R_1 (R_2 + R_L) + R_2 R_L}{R_2 (1 + a) + R_L}$$

ii) 6050 find R_o if $r_o \neq 0$ ($\hat{=}$ a $\neq \infty$)

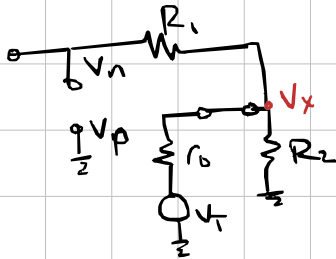
r_o :



No current through R_1 !

$$r_o = R_2 + r_o$$

T_{sc} :

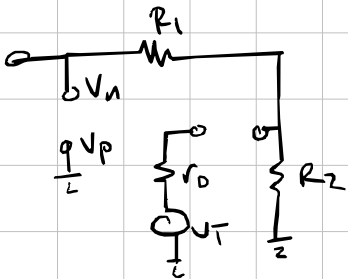


$V_n = V_x$ (no current through R_1)

$$V_x = V_T \frac{R_2}{R_2 + r_o}$$

$$V_o = -V_n = -V_x \quad T_{sc} = a \frac{R_2}{R_2 + r_o}$$

T_{oc} :



No connection between V_n & V_T !

$$V_n = 0$$

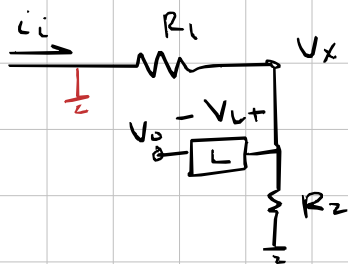
$$T_{oc} = 0$$

$$R_o = R_2 + r_o \left(1 + a \frac{R_2}{R_2 + r_o} \right)$$

$$R_o = r_o + R_2(1+a)$$

if $r_o = 0 \rightarrow R_o = R_2(1+a)$ ✓

iii) voltage compliance doesn't need to be exact, so we can assume $a \rightarrow \infty$ (for 6050)



$$V_x = -i_i R_1$$

$$V_o + V_L = V_x = -i_i R_1$$

$$V_o = -i_i R_1 - V_L$$

$$-V_{sat} \leq V_o \leq V_{sat}$$

$$-V_{sat} \leq -i_i R_1 - V_L \leq V_{sat}$$

$$V_{sat} \geq i_i R_1 + V_L \geq -V_{sat}$$

$$-V_{sat} - i_i R_1 \leq V_L \leq V_{sat} - i_i R_1$$

iv) with $R_2 = 200 \Omega$ & $i_i = -9.5 \text{ mA}$: $V_x = -9.5 \text{ mA} (200 \Omega)$
 $V_x = -1.9 \text{ V}$

with $V_L = 5 \text{ V}$, then $V_o = V_x - V_L = -6.9 \text{ V}$

then power consumed to produce output is:

$$P_o = (V_o - V_{p-}) i_o = (-6.9 + 12 \text{ V}) 10 \text{ mA}$$

$$P_o = 51 \text{ mW}$$

Quiescent power: $P_Q: (12 - 12) 1 \text{ mA} = 24 \text{ mW}$

$$P_{\text{Total}} = 75 \text{ mW}$$

v) $i_i = i_o + i_2 = 10 \text{ mA} - 9.5 \text{ mA} \rightarrow i_i = 0.5 \text{ mA}$

$$0 - V_x = i_i R_1$$

$$R_1 = -1.9 \text{ V} / 0.5 \text{ mA}$$

$$R_1 = 3.8 \text{ k}\Omega$$

OR $A = \left(1 + \frac{R_1}{R_2}\right) = \frac{10 \text{ mA}}{0.5 \text{ mA}} = 20$

$$\frac{R_1}{R_2} = 19 \rightarrow R_1 = 3.8 \text{ k}\Omega$$